

PatientTriage.ai

“Triage is a snapshot. Risk isn’t.”

Active Autonomous Emergency Department Safety Control Tower • Accenture Innovation Challenge 2026

COMPETITION TRACK	CORE STACK	VERIFICATION	REPOSITORY
Round 2 Technical Prototype	Python 3.13 / FastAPI / React 19	51/51 Tests Passed (100%)	github.com/freya1705/PatientTriageAI

1. Project Overview & Control Tower Concept

PatientTriage.ai is an active autonomous emergency department clinical decision-support control tower. Traditional emergency triage operates as a static, one-time snapshot at intake. PatientTriage.ai continuously tracks vital sign trajectory velocity (ΔSpO_2 , ΔHR), observation shelf-life (Safety Clocks), data uncertainty (Unknown $\neq$ Safe), and clinician attention allocation (The Attention Gap). It provides continuous decision support, while deterministic guardrails protect patient safety and clinicians retain 100% decision authority.

The system continuously answers: 1. Who needs attention right now? • 2. Why are they becoming unsafe? • 3. What is the exact next clinical action?

2. The Core Problem & Competitive Moat (Vs. Legacy EHR Scores)

- 1. Silent Waiting Room Deterioration:** ESI Level 3/4 patients worsen unmonitored; physiological decline may remain undetected until a subsequent reassessment or clinical deterioration becomes apparent.
- 2. Missing Vitals & Stale Data:** When vitals are missing, legacy systems default to low urgency. Under *Unknown* is *NOT Safe*, missing data heightens clinical vigilance.
- 3. The Attention Bottleneck:** Attended critical patients block static queues, while unattended deteriorating patients remain buried.
- Competitive Moat vs. Native EHRs (Epic EDI / Cerner MEWS):** Legacy EHR algorithms are designed for admitted inpatients in hospital beds and only score raw severity. PatientTriage.ai is purpose-built for the waiting room and uniquely factors in **physician coverage** and **evidence staleness decay**.

3. System Architecture & 8 Core Intelligence Engines

Tier	Responsibilities & Scope	Key Modules
Tier 1: Deterministic Guardrails	Deterministic red-flags ($\text{SpO}_2 < 85\%$, $\text{SBP} < 75 \text{ mmHg}$, FAST Stroke, pediatric stridor) that bypass statistical models. Downgrade safety blocking.	safety_guardrails.py downgrade_guard.py
Tier 2: AI & Trajectory Engine	Continuous vital velocity (ΔSpO_2 , ΔHR), confidence decay, uncertainty scoring, Attention Gap re-ranking, 108 EMS Pre-Arrival Telemetry with LOINC FHIR parser, and Referral Eligibility Scoring (RES).	risk_engine.py attention_gap_engine.py ems_routes.py
Tier 3: Multi-Surface Dispatch	4 Workspaces: Overview (Control Tower), Nurse Tasks ('Next 5 Mins'), Floor Map (Chairs 1–20), and Lab Orders (Standing Pre-Orders). Unified slide-over Patient Drawer.	PatientDrawer.jsx EDPressureMap.jsx

4. Core Intelligence Engines & Attention Gap Formula

The Attention Gap Priority Equation:

Action Priority Score = $(w_r \times \text{Risk} + \text{Urgency}) + (w_d \times \text{Deterioration}) + (w_s \times \text{Staleness}) + \text{Wait Hazard} + (w_u \times \text{Uncertainty}) - (w_c \times \text{Clinical Coverage})$

Parameter Calibration: $w_r = 1.0$ (base risk 0–100) • $w_d = +25$ to $+40$ pts ($\Delta\text{SpO}_2 \leq -5\%$ or $\Delta\text{HR} \geq +20 \text{ bpm}$) • $w_s = +20$ to $+35$ pts (safety expiry) • $w_u = +15$ to $+25$ pts (missing vitals) • $w_c = -35$ pts (discounts attended patients, elevating unattended cases to Rank #1).

5. Benchmark Cohort & Empirical Impact Evaluation

Evaluated across 20 synthetic clinical scenarios and verified across 51 automated backend pytest cases:

Performance Dimension	Traditional Static Triage	PatientTriage.ai	Benchmark Impact
Waiting Deterioration Catch Rate	0/20 detected	20/20 synthetic scenarios detected	100% Benchmark Coverage
Stale Observation Flagging	0/20 flagged	20/20 synthetic cases flagged (EXPIRED)	Zero Unmonitored Stale Waits
False Reassurance on Missing Vitals	High (Treated Normal)	0% False Reassurance (Unknown ≠ Safe)	Eliminates Under-Triage
Attention Gap Optimization	None (Attended Block)	Active (Elevates Unattended)	Optimized Clinician Utilization
Pre-Arrival Telemetry (108 EMS)	Manual Handover Delay	Automated FHIR LOINC Ingestion	1-Click Bay Pre-Allocation
Unsafe Priority Downgrades Blocked	0 Guardrails	100% Guarded (Proof Required)	100% Downgrade Guarded

**Note: Results reflect simulated evaluations across 20 synthetic clinical benchmark scenarios for prototype demonstration.*

6. Phased Implementation Roadmap & Precise Trial Endpoints

- **Phase 1 (Q3 2026):** Lab Validation • 20 synthetic scenarios verified across 51 automated tests; sub-15ms inference latency.
- **Phase 2 (Q4 2026):** Shadow Clinical Trial • Silent background deployment with Epic/Cerner via HL7 FHIR and CDS Hooks.
- **Phase 3 (Q1–Q2 2027):** Live Pilot • **Primary Safety Endpoint:** Reduction in Mean Time to Escalation (MTTE) (>45% faster); **Operational Endpoint:** Nurse false-alarm rate < 2 alerts/nurse/shift; **Economic Endpoint:** ≥25% reduction in Left-Without-Being-Seen (LWBS).
- **Phase 4 (Q3 2027+):** Multi-Hospital Enterprise Network Scaling • Regional dashboarding, telemedicine, and referral diversion.

7. Quick Start & Security Architecture

Launch: .\start.ps1 | Run 51 Automated Tests: python -m pytest -v | Web UI: http://localhost:5173

• **Zero PHI:** 100% synthetic physiological datasets • **Air-Gapped:** Zero external cloud LLM calls • **Audit Ledger:** Append-only event logs.

Regulatory & Safety Notice: PatientTriage.ai is a clinical decision-support research prototype developed for the Accenture Innovation Challenge 2026. All patient cohorts are synthetically generated. This system is positioned under FDA Non-Device CDS (21 U.S.C. § 360aaa-1) and licensed clinicians retain 100% final override authority.